

Characterisation and Quantification of Data Centre Flexibility for Power System Support

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Highlights

- Linked annual optimisation preserves physical and workload continuity.
- Integrated operation cuts annual electricity cost by 5.049%.
- Flexible workload has greater value than UPS or TES capacity.
- All year-end deferred work is completed within its deadline.
- Terminal-value treatment changes annual cost by at most 0.000381%.

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Abstract—The rapid growth of data centres creates both a power-system challenge and an opportunity for demand-side flexibility. This paper develops a linked rolling-horizon optimisation of IT workload scheduling, uninterruptible power supply (UPS) dispatch, cooling dynamics and thermal energy storage (TES) for a 1 MW data centre. Unlike independent daily studies, physical states and unfinished workload cohorts are carried exactly between local-day horizons across a complete year, including daylight-saving transitions. Against a workload-at-arrival, storage-disabled benchmark with cost-optimised cooling, integrated operation reduces 2025 electricity cost from GBP 550,503.59 to GBP 522,707.23, a saving of 5.049%. Full-year one-at-a-time endpoints show that the flexible-workload share has the largest effect on the saving, followed by UPS energy capacity and then TES capacity. Representative-week intermediate points preserve this response ranking but overstate the absolute annual effect and are therefore used only as centred response-shape evidence. All continuity, workload, exclusivity and settlement-reconciliation audits pass. At the year boundary, 0.642875 CPU-hours outstanding at midnight are completed within deadline in the actual 1 January 2026 look-ahead, while alternative terminal energy values change annual cost by at most 0.000381%. The results demonstrate the annual economic value of coordinating data-centre flexibility resources and establish an audited baseline for a subsequent duration-aware grid-service assessment.

Index Terms—Data Centre, Power system flexibility, Demand side flexibility, Load Shifting, Thermal Inertia

I. INTRODUCTION

GLOBAL digitalisation, propelled by artificial intelligence, is driving the rapid expansion of data centres, whose electricity demand is both substantial and spatially concentrated. This growth coincides with two other major shifts in the energy landscape: the increasing share of variable renewable generation needed for decarbonisation, and the widespread adoption of new demand-side technologies such as electric vehicles and heat pumps [1], [2]. The combined effect places significant strain on modern power systems by intensifying peak loads, steepening net load ramps, and aggravating local network constraints. Consequently, there is

an escalating need for power system flexibility to maintain the crucial balance between electricity supply and demand.

According to the IEA, global data centre electricity consumption was around 415 TWh in 2024, with projections suggesting this will exceed 945 TWh by 2030, raising their share of global electricity demand from approximately 1.5% to 3% [3]. This expansion is not uniform, with over 80% of the increase expected to originate from the United States and China [3]. Moreover, geographical clustering intensifies the impact on local grids. For instance, data centre consumption already accounts for over 20% of national demand in Ireland, and 25% in Virginia, USA [3]. In Great Britain, annual electricity consumption by data centres is forecast by National Grid ESO to rise from 7.6 TWh in 2024 to between 30–71 TWh by 2050 [4].

To manage the volatility introduced by these trends, power systems require a substantial increase in flexibility services. Traditionally provided by dispatchable fossil fuel generators, this solution is becoming obsolete under decarbonisation mandates. The JRC [5] projects that the European Union's flexibility requirement will rise to 24% of total electricity demand in TWh by 2030 and to 30% by 2050. Similarly, Great Britain's Clean Flexibility Roadmap targets an expansion of flexibility capacity from 25.2 GW to between 54–66 GW by 2030 [6]. Demand-side flexibility, where consumers actively modulate their electricity use to support the grid, presents a key solution.

While their growth contributes to the challenge, data centres are equipped to provide flexibility. Their shiftable IT workloads, uninterruptible power supply (UPS) systems with battery storage, and inherent thermal inertia as well as cold storage, can be leveraged to modulate power consumption without disrupting core services. Each of these internal assets possesses a distinct flexibility profile in terms of magnitude, duration, ramp rate, and recovery time [7]. For example, UPS batteries can offer a near instantaneous response, whereas thermal systems are characterised by lower ramp rates, providing

a more gradual adjustment [7], [8]. Data centres therefore present a dichotomy, they are both a growing strain on the power system and a potentially crucial asset for grid flexibility.

In the context of power systems, flexibility is defined as the capability of maintaining the balance between generation and demand across all time horizons, by adjusting either energy production or consumption in response to sudden variations [1], [2], [4], [9]. Upward flexibility is required when demand exceeds generation and involves a DC reducing its power consumption from the grid. Conversely, downward flexibility is needed when generation surpasses demand, and is provided by a DC increasing its electricity consumption to absorb the surplus energy [1], [10]. In this study, a data centre's flexibility is quantified by its capacity for power consumption deviation relative to a baseline, and the maximum duration for which this deviation can be sustained.

This paper addresses these challenges with a linked annual optimisation of IT workload, UPS dispatch and cooling operation against signed electricity prices. Its principal contribution is an auditable rolling-horizon formulation that passes physical states and unfinished workload between local days instead of resetting the facility at midnight. The study quantifies annual cost value, tests full-year sensitivity endpoints for the three internal flexibility resources, and defines a recovery-constrained duration assessment to be applied to an objectively selected representative day.

The main contributions are therefore: (1) a tractable integrated data-centre model that preserves the relevant coupling between IT execution, UPS operation, cooling power and TES state; (2) a complete 2025 rolling chain with explicit state, workload, solver and settlement audits; and (3) annual sensitivity evidence that separates full-year effect sizes from representative-week response shapes.

The remainder of this paper reviews the literature, defines the linked annual method and benchmark, presents the completed annual results and sensitivities, and identifies the deferred representative-day flexibility assessment.

A. Data Centre Architecture

DCs are integrated facilities housing IT equipment, such as servers and network devices, which are supported by critical power and cooling infrastructure. As shown in Figure 1, electrical continuity is maintained by uninterruptible power supplies (UPS) with associated battery systems, while cooling systems, including chillers, Thermal Energy Storage (TES) tanks and Computer Room Air Conditioning (CRAC) units, manage the substantial heat generated by the servers. Thermal efficiency is often enhanced through optimised airflow strategies like the hot-aisle/cold-aisle arrangement.

Each of these core subsystems presents an opportunity for providing demand-side flexibility to the power grid. The strategic scheduling of IT workloads, the dispatch of UPS battery storage, and the leveraging of the TES tank and the facility's inherent thermal inertia allow a DC to modulate its power consumption. The DC assets are highly interdependent, requiring an integrated approach that considers all of them simultaneously. This is because a change in one subsystem

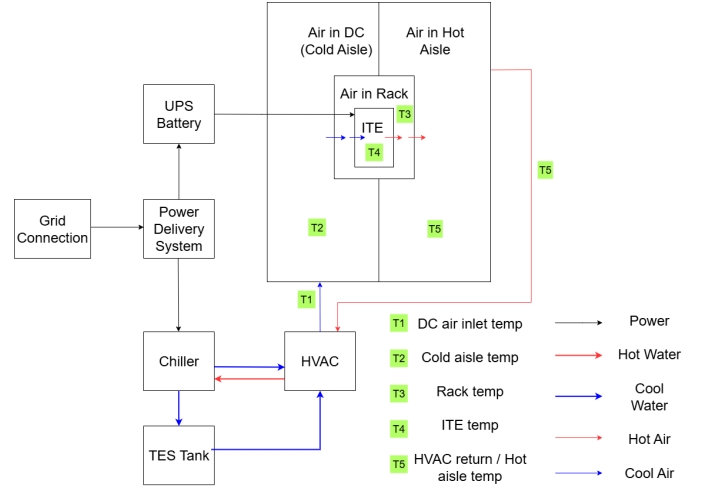


Fig. 1. Data Centre Layout

changes the feasible operating range of the others: deferring IT workload reduces both server power and subsequent heat generation, cooling actions are constrained by thermal states and the TES charge level, and UPS discharge can support IT execution while altering the grid power draw. A component-by-component assessment can therefore overestimate flexibility if it ignores recovery requirements, or underestimate it if it misses compensating actions across assets. Consequently, DCs can transition from being passive electrical loads to active assets that enhance the resilience and flexibility of the power system.

B. IT Workload Background

The fundamental role of a DC is to execute IT workloads, which encompass all computational and data processing tasks. These workloads can be broadly classified based on their time sensitivity, as defined by service-level agreements (SLAs) [11], [12]. Inflexible workloads, also known as interactive workloads, are latency-critical and require immediate execution to ensure quality of service for user-facing applications like real-time streaming or transaction processing.

In contrast, flexible IT workloads, often referred to as batch workloads, can tolerate execution delays within predefined time windows without impacting service quality. These typically non-user-facing tasks, such as periodic data analytics, backups, or machine learning model training, can be deferred or rescheduled. This ability to shift the timing of their execution is the primary source of IT-based flexibility, allowing the DC to adjust its computational power draw in response to external signals like electricity prices or grid operational needs [12].

II. LITERATURE REVIEW

Data centre flexibility has attracted growing attention as an enabler of demand-side participation in power systems, with a growing body of literature modelling flexibility at the asset level, predominantly focusing on IT workloads and, to a lesser extent, UPS batteries, cooling systems and other subsystems.

A smaller body of work has attempted integrated models that capture the joint flexibility of multiple DC assets. This review therefore first presents the flexibility modelling approaches found in the literature for each DC asset, and then discusses studies that move towards an integrated data centre flexibility assessment.

IT workload is the most fundamental component of a DC and is the focus for many flexibility studies. One study develops a machine learning method to predict DC energy consumption and server temperature [13]. These data are then used to optimise the scheduling of IT workloads to minimise fuel costs and power utilisation at times of high grid carbon intensity. On average, the implementation achieved a 6.5% load reduction for a winter day during periods of high CO₂ intensity. Radovanović et al. [14], performed predictive modelling of grid carbon intensity, trained day-ahead demand prediction models and used them to optimise IT workload for minimum carbon emissions. Cao et al. [15] propose and validate a data-driven methodology using real-world Alibaba trace data to assess the power flexibility potential of Internet Data Centres (IDCs) by rescheduling periodic IT workloads. Their findings quantify significant flexibility, particularly in early morning hours, revealing potential upward/downward load shifts of 400kW and 325kW, respectively—corresponding to approximately 50% and 40.6% of the facility’s peak demand—by optimising job schedules based on power system needs and renewable energy availability. In this study, IT workloads are defined as either flexible or inflexible, with flexibility here referring to flexibility in time.

UPS batteries are currently providing power flexibility to the grid in numerous DCs around the world and have been shown to be successful [16]–[18]. Lithium-ion batteries, which are increasingly the dominant form used for UPS, are capable of rapidly providing or consuming power when needed [8]. These characteristics make them well suited to frequency regulation, a service many power providers require. Results from [8] found UPS batteries could be used within their normal operating range to provide the required services to the grid and in doing so generate a reasonable revenue stream. Furthermore, the battery lifetime would not be affected, and the approach does not require additional cost.

Cooling makes up a large proportion, as high as 40%, of DC energy consumption [19]. The American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE) recommends maintaining an indoor temperature of 18°C to 27°C to ensure optimal DC performance [20]. Exploiting the thermal inertia provided by this temperature range has been shown to enable power consumption flexibility in DCs [21] [22]. Thermal Energy Storage (TES) systems can enhance cooling flexibility by decoupling the timing of heat removal from the cooling generation process. Du et al. [23] propose and evaluate a framework to enhance the energy flexibility of district heating (DH) systems integrated with DC waste heat recovery. The study utilises dual short-term TES tanks to enable simultaneous peak shaving (demand-driven) and load shifting (price-driven) within the hybrid system. Simulation results demonstrated that this dual TES approach significantly improved flexibility, achieving up to a 10% peak demand

reduction and a 2.1% load shift, leading to a 3.2% operational cost saving in the case study. A ‘synergistic control strategy for data centre frequency regulation which uses both IT and cooling systems’ has also been proposed [24]. Results found that a revenue saving of 4% could be made by implementing the proposed strategy. Once depleted, the thermal inertia of the DC atmosphere / TES must be recovered to baseline levels. The recovery time taken to do so will affect how the flexibility of the asset can be utilised.

While individual assets offer flexibility, the flexibility characteristics change when these assets are integrated into one system. Maximising the overall potential requires coordinating multiple resources and evaluating their diverse characteristics, interactions, and constraints. One study proposes a co-optimisation of the IT workload and cooling system, recognising the strong operational link between them [24]. The work by S Xiang et al. [25] identifies that data centre optimisation models often ignore IT equipment constraints like start-stop conditions and ramp rates, which can cause excessive wear. The authors propose a model integrating these constraints with facility thermal dynamics, solved using a Deep Deterministic Policy Gradient (DDPG) algorithm. This approach was then validated for practicality and efficiency using a large-scale simulation of a 100,000-device data centre. H. Xu et al. [26] propose an optimisation which integrates IT servers, cooling infrastructure, energy storage, generators and renewable energy, and aims to maximise power demand flexibility. The objective function is the cumulative change in energy taken from the grid before and after the optimisation was applied. Other studies use a similar approach, but instead minimise the difference between DC power consumption and the desired power consumption of the power supply operator [27].

While individual DC assets, including IT workloads, UPS batteries and cooling systems, have each been studied for their flexibility potential, the interplay between these assets and its impact on the overall flexibility of the DC remains largely unexplored. Most existing studies adopt a DC-centric, cost-minimisation perspective and do not quantify the magnitude and duration of grid-facing flexibility that can be delivered once coupled asset constraints and post-event recovery are accounted for. As DCs scale towards the gigawatt level, this power-system-centric perspective becomes increasingly important, because system operators require duration-certified flexibility offers and a clear understanding of which assets enable them; this perspective remains largely absent from the literature.

Addressing these gaps, this study first establishes an audited annual operating baseline and defines a recovery-constrained flexibility analysis that can determine, for a given start time and power deviation, the maximum feasible duration. The annual evidence is completed here. The refreshed duration grid is reserved for the next revision so that it can be solved from an exact annual checkpoint rather than reusing results from an independent-day formulation. This separation is important because magnitude and duration are meaningful to the power system only when the baseline state and post-event recovery are internally consistent.

III. METHODOLOGY

A. Rolling-horizon formulation

The data centre is modelled at 15-minute resolution as an integrated mixed-integer linear programme (MILP). Each local calendar day forms the committed core of one solve and is followed by a three-hour look-ahead. The core therefore contains 92, 96 or 100 intervals at the daylight-saving transitions, while the look-ahead always contains 12 intervals. Only core intervals are included in the reported calendar-year cost. The look-ahead establishes a feasible continuation for workload and physical states.

The state at the end of a committed core,

$$\mathbf{x}_d^{\text{end}} = (E^{\text{UPS}}, E^{\text{TES}}, T_{\text{IT}}, T_{\text{R}}, T_{\text{CA}}, T_{\text{HA}}), \quad (1)$$

is imposed exactly as the initial state of day $d+1$. Outstanding flexible workload is passed in the same way as timestamped cohorts with absolute deadlines. This linking prevents the artificial daily reset of stored energy, temperature or deferred work.

B. IT workload and electrical power

The workload trace specifies inflexible CPU utilisation and four flexible tranches whose maximum delays are 0.5, 1, 2 and 3 hours. If a_c is the CPU-hours in cohort c and $\mathcal{W}_c$ is its feasible execution window, the scheduled fraction $u_{c,t}$ satisfies

$$\sum_{t \in \mathcal{W}_c} u_{c,t} \Delta t = a_c, \quad (2)$$

$$u_t^{\text{inflex}} + \sum_c u_{c,t} \leq 1. \quad (3)$$

The first equation is enforced within the available horizon when the deadline is visible; otherwise the unexecuted balance is carried forward. The complete three-hour look-ahead is long enough to expose every deadline generated in a core.

IT power is a convex nonlinear function of aggregate CPU utilisation in the source equipment model. It is represented by a four-segment non-uniform DLOG piecewise-linear approximation. Across the feasible utilisation range, the maximum approximation error is 6.241 kW, or 0.624% of the 1 MW IT rating.

C. UPS, cooling and thermal states

The UPS energy recursion is

$$E_{t+1}^{\text{UPS}} = E_t^{\text{UPS}} + \eta_{\text{ch}} P_{\text{ch},t}^{\text{UPS}} \Delta t - \frac{P_{\text{dis},t}^{\text{UPS}}}{\eta_{\text{dis}}} \Delta t. \quad (4)$$

Binary mode variables prevent simultaneous charging and discharging. The central UPS has a 600 kWh energy capacity and a 50% lower state-of-charge bound. The first annual interval starts at 600 kWh; thereafter no daily cyclic condition is imposed.

The chiller supplies the CRAC directly and can also charge the 1000 kWh-th TES. The TES may discharge to the CRAC, with mutually exclusive charge and discharge modes. TES begins at 500 kWh-th and uses 0.9 charge and discharge efficiencies. The four-node facility thermal model is discretised

with backward (implicit) Euler boundary states. Both benchmark and optimised cases use the same 18–23 °C cold-aisle range. A conservative operating rule,

$$Q_t^{\text{cool}} \geq P_t^{\text{IT}}, \quad (5)$$

requires contemporaneous cooling provision to be at least the IT heat load. This avoids treating unvalidated heat accumulation as a flexibility resource. Outdoor temperature is fixed at 22 °C and the chiller coefficient of performance at 5 throughout the stylised annual experiment.

Total grid import is

$$P_t^{\text{grid}} = P_t^{\text{grid,IT}} + P_{\text{ch},t}^{\text{UPS}} + P_t^{\text{ch,CRAC}} + P_t^{\text{ch,TES}} + P^{\text{OD}}, \quad (6)$$

where P^{OD} is the constant auxiliary load. The objective minimises the signed settlement cost over the complete core and look-ahead:

$$\min \sum_t \pi_t P_t^{\text{grid}} \Delta t / 1000. \quad (7)$$

Negative prices are retained. The central model assigns no storage degradation cost, standing loss, additional UPS reserve or terminal-energy credit.

D. Annual cases and sensitivity design

The annual benchmark executes workload at arrival and disables UPS and TES dispatch, while retaining cost-optimised cooling within the same temperature bounds. The optimised case activates flexible workload scheduling and both storage systems. This is a comparison of operational flexibility with the implemented benchmark; it is not a comparison against a fixed 22.5 °C cooling set point.

The input price is the actual signed 2025 Intermittent Market Reference Price (IMRP) series, expanded from hourly to quarter-hour resolution. The local-clock workload profile repeats daily, while an absolute UTC timeline preserves both daylight-saving transitions. Pyomo and HiGHS are used with a requested relative MILP gap of 0.1%, a 300-second daily limit and a normal accepted gap of 1%.

Full-year one-at-a-time endpoints vary flexible-workload share, UPS energy capacity and TES energy capacity to $0.5\times$ and $1.5\times$ their central values. Storage power ratings are held fixed and starting stored-energy fractions are preserved. Twelve representative weeks provide the intermediate $0.75\times$ and $1.25\times$ response shape. Because the sampled central saving differs from the annual saving, sampled curves are centred on their own central case and are not reported as annual estimates.

E. Representative-day flexibility assessment

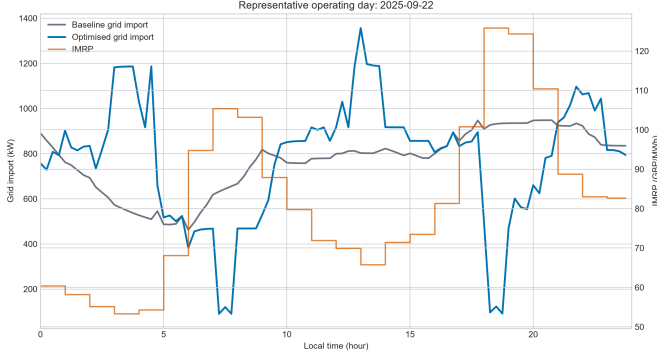
The duration-aware flexibility formulation evaluates a request beginning at t_0 with grid deviation ΔP from the annual optimised baseline. During the event,

$$\left| P_t^{\text{grid}} - \left(P_t^{\text{grid,base}} + \Delta P \right) \right| \leq \epsilon_P, \quad (8)$$

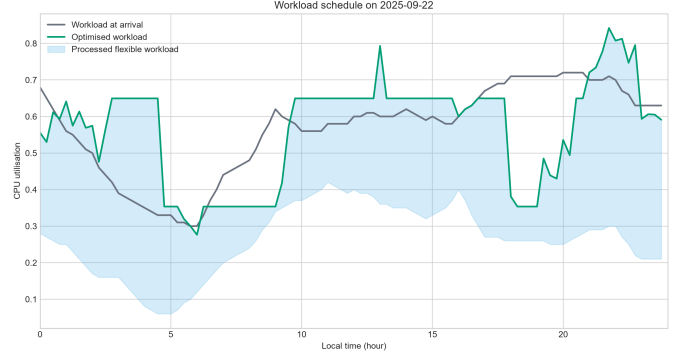
and a binary search finds the longest feasible duration. Recovery constraints require the terminal UPS, TES and four thermal states to return to the baseline continuation state. An objective

Table I
FULL-YEAR OPERATING-COST AND SENSITIVITY RESULTS.

Case	Cost (GBP)	Saving (%)	Change (pp)	Energy (GWh)
Baseline	550,504	0.000	–	6.718
Flexible workload 0.5×	531,117	3.522	-1.528	6.782
Flexible workload 1.0×	522,707	5.049	+0.000	6.794
Flexible workload 1.5×	516,058	6.257	+1.208	6.807
UPS capacity 0.5×	525,374	4.565	-0.484	6.771
UPS capacity 1.5×	520,222	5.501	+0.452	6.815
TES capacity 0.5×	524,080	4.800	-0.249	6.784
TES capacity 1.5×	522,057	5.167	+0.118	6.799



(a) Grid import and IMRP price.



(b) Workload execution.

Fig. 2. Annual baseline and optimised operation on the objectively selected representative day, 22 September 2025.

representative day is selected by the smallest standardised distance to annual price, grid, CPU, cost and opening-state statistics; this procedure selects 22 September 2025.

Draft status: the refreshed Scenario 3 solve has deliberately been deferred. No duration-envelope values from the obsolete independent-day model are carried into the annual results below. The prepared runner reconstructs the selected day from its audited annual checkpoint and will populate the flexibility heatmap and component decompositions in a subsequent revision.

IV. ANNUAL RESULTS AND DISCUSSION

A. Central annual result

The workload-at-arrival benchmark costs GBP 550,503.59 in 2025, whereas the integrated optimisation costs GBP 522,707.23. The resulting saving is GBP 27,796.37, or 5.049%. Table I gives the central case and all completed full-year sensitivity endpoints.

The optimised case imports 6.794 GWh compared with 6.718 GWh in the benchmark. The economic benefit therefore arises from moving electrical consumption to lower-priced intervals, not from reducing annual electricity use. The 1 MW facility's annual peak import rises from 974.9 kW to 1681.9 kW as storage charging and deferred-work recovery are concentrated in favourable intervals. Consequently, the result should be interpreted as an energy-market operating cost benefit; network-capacity charges or an explicit grid-import limit would change the dispatch.

Figures 2 and 3 illustrate how flexible work, UPS dispatch and cooling are coordinated on the selected day. They are

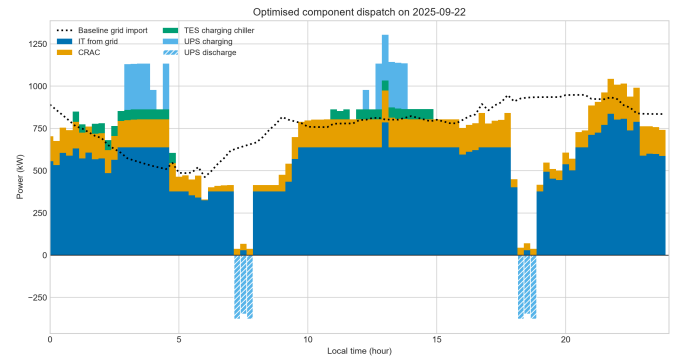


Fig. 3. Optimised component dispatch on 22 September 2025. Negative bars denote UPS discharge; the dotted line is benchmark grid import.

descriptive slices through the linked annual trajectory, not independent daily optimisations: opening storage, temperature and workload states are taken directly from the 21 September checkpoint.

B. Sensitivity and asset ranking

Flexible-workload share is the dominant tested internal parameter. Halving it reduces the annual saving to 3.522%, while increasing it by half raises the saving to 6.257%. The corresponding changes from the central result are -1.528 and +1.208 percentage points. UPS capacity ranks second: its endpoint range is 4.565–5.501%. TES capacity has the smallest effect, with a 4.800–5.167% range. Thus the frozen

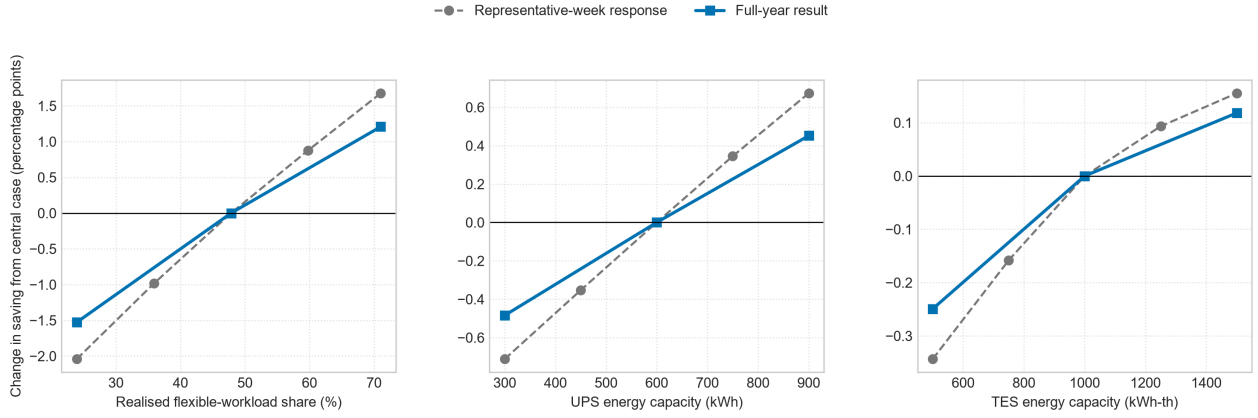


Fig. 4. Sensitivity response relative to the respective central case. Solid blue endpoints are full-year runs; dashed grey intermediate curves are from twelve representative weeks and show response shape only.

ranking is flexible workload, UPS energy capacity, then TES energy capacity.

Figure 4 combines the full-year endpoints with the centred representative-week curves. The sampled central saving is 6.713%, compared with 5.049% annually, so the sampled absolute percentages would overstate annual effects. Their use here is limited to confirming direction, curvature and the same parameter ranking. No interaction effects are inferred from the one-at-a-time design.

C. Continuity, solver quality and terminal treatment

Every annual chain passes timestamp coverage, state hand-off, workload conservation, charge/discharge exclusivity and settlement-cost reconciliation audits. The few time-limited daily solves retain feasible incumbents. The largest exceptional relative gap occurs in the low-flexibility case on 25 May: a 900-second rerun leaves a 5.652% relative gap because the objective is close to zero and negative, but the absolute incumbent-bound width is only GBP 0.102. The committed cost changes by approximately GBP 0.002, the downstream state is unchanged and every physical audit passes. It is retained with this explicit qualification.

At midnight after 31 December, 0.642875 CPU-hours from the central core remains within deadline. It is fully completed in the 1 January 2026 look-ahead, which uses the available actual IMRP values of GBP 46.49, 40.19 and 28.99/MWh for the first three settlement periods. Calendar-year reporting includes only 2025 committed intervals; the look-ahead is a feasibility certificate and includes ordinary 2026 demand that cannot be attributed to 2025.

The central convention assigns zero value to terminal UPS and TES energy. A continuation-value test spanning the median to the maximum observed 1 January price changes the annual cost by at most 0.000381%. At the maximum tested value, the final horizon retains 600 kWh in the UPS and 1000 kWh-th in TES instead of ending at their lower states. This negligible annual effect supports the zero-value central convention while making its implication transparent.

D. Scope and limitations

The analysis is a stylised annual operating experiment rather than a forecast for a specific facility. Ambient temperature, COP and the local-clock workload shape are fixed, storage ageing and standing losses are omitted, and network tariffs and service revenues are outside scope. The conservative $Q^{\text{cool}} \geq P^{\text{IT}}$ rule also limits the use of building thermal mass. The cost results establish the value and relative importance of the modelled resources under signed energy prices; they do not alone establish the profitability of a commercial flexibility product.

V. CONCLUSION

Linked rolling optimisation across all of 2025 reduces the model data centre's electricity cost from GBP 550,503.59 to GBP 522,707.23, a saving of 5.049%. The calculation preserves physical and workload continuity across every local day, including daylight-saving transitions, and reconciles directly to signed IMRP settlement prices.

Full-year endpoints show that workload flexibility is the most influential tested internal resource, followed by UPS energy capacity and then TES energy capacity. Representative-week intermediate points preserve this ranking and response shape but overstate the absolute annual effect, so they are used only as centred comparative evidence. Year-end continuation-value tests alter annual cost by no more than 0.000381%, and all 0.642875 CPU-hours outstanding at midnight are completed within deadline in the actual 1 January look-ahead.

The next revision will apply the prepared duration-aware flexibility runner to 22 September 2025 using the exact annual checkpoint state. That study will add recovery-certified magnitude-duration envelopes without mixing results from the superseded independent-day formulation. Further work should incorporate weather-varying cooling performance, storage degradation, network constraints, service revenues and factorial interactions between resources.

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ACKNOWLEDGEMENTS

This work was supported by the Engineering and Physical Sciences Research Council (EPSRC) and the Economic and Social Research Council (ESRC) through funding provided to the Energy Demand Research Centre (grant number EP/Y010078/1).

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